Arnav Tyagi

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LinkedIn | *GitHub* | *HackerRank*

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript
AI & Generative AI: Machine Learning, Deep Learning, LLMs, Generative AI, Prompt Engineering, RAG
AI Frameworks & Tools: LangChain, LCEL, Mistral AI, ChromaDB, Hugging Face
Backend Development: Django, DRF, FastAPI, REST APIs, Pydantic
Databases: MongoDB, SQL, SQLite
Data & ML Libraries: NumPy, Pandas, Scikit-learn, TensorFlow
Developer Tools: Git, GitHub, Postman, pytest, uv, Docker
Cloud & Deployment: AWS, Render

PROJECTS

LangChain Docs Q&A – Retrieval-Augmented Generation System [Live Demo]
Python, LangChain (LCEL), Mistral AI, ChromaDB, FastAPI, Streamlit, pytest, uv
[GitHub]

- Built and deployed a RAG system over 7 official LangChain documentation files (∼316 chunks), generating context-grounded answers with source citations.
- Designed provider-isolated LLM and embedding factories, and implemented MMR-based retrieval with strict grounding prompts and an insufficient-context fallback.
- Exposed the pipeline through FastAPI REST endpoints, a Typer CLI, and a Streamlit interface, using a shared service layer and Pydantic API validation.
- Evaluated 3 chunking configurations and reported Recall@4 of 1.00 and keyword coverage above 0.85; wrote 7 pytest tests covering ingestion, chunking, and retrieval.

MediConnect AI – Healthcare & Disease Screening Platform [GitHub]
Python, Django, DRF, TensorFlow/Keras, scikit-learn, SQLite, HTML/CSS/JavaScript

- Built a multi-role Django platform (patients, doctors, admins) with authentication, doctor discovery, disease screening, patient records, and appointment booking.
- Integrated 4 ML screening workflows (symptom-based, diabetes, heart disease, pneumonia) using scikit-learn and TensorFlow/Keras, achieving 78–92% accuracy.
- Developed a Django REST Framework API linking predictions to doctor recommendations and appointments; added one-to-one booking constraints and PDF report generation.

SmartAgri – Agriculture Decision Support Platform [Live Demo]
Python, Django, SQLite, REST APIs, HTML, CSS, JavaScript

- Developed a Django-based agriculture platform with authentication and crop management features.
- Integrated OpenWeather and Government APMC APIs to deliver real-time weather insights and mandi price updates for informed crop planning.
- Designed relational database models using Django ORM and SQLite for managing user profiles and crop records.
- Deployed the application on Render using Gunicorn and environment variables for production-ready deployment.

EDUCATION

Bachelor of Technology (B.Tech.) in Computer Science & Engineering Aug 2023 – Aug 2027
Dr. A.P.J. Abdul Kalam Technical University

Class XII (CBSE): 90% Vivekananda School, Delhi
Class X (ICSE): 92% Holy Angels School, Ghaziabad

CERTIFICATIONS

- Full Stack Web Development with AI Tools — EY, AICTE & Edunet Foundation
- EY AI Skills Passport — EY & Microsoft
- AWS Academy Graduate – Cloud Foundations — Amazon Web Service
- Blockchain Fundamentals Certification — Chainlink Labs
- MongoDB Atlas Search — MongoDB
- Java Foundations — Oracle